1. **Loaded data from csv:**

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("ecommerce_sales.csv")

display(df)
```

	order_id	order_date	month	products	region	units	unitprice	discount
0	ORD001	2026-01-05	2026-01	Laptop	South	2	55000	2000
1	ORD002	2026-01-08	2026-01	Smartphone	North	5	22000	1500
2	ORD003	2026-01-12	2026-01	Headphones	East	10	2500	500
3	ORD004	2026-02-03	2026-02	Laptop	West	1	55000	1000
4	ORD005	2026-02-09	2026-02	Smartphone	South	8	22000	3000
5	ORD006	2026-02-14	2026-02	Keyboard	North	12	1500	600
6	ORD007	2026-02-20	2026-02	Headphones	West	15	2500	750
7	ORD008	2026-03-02	2026-03	Smartphone	East	6	22000	1800
8	ORD009	2026-03-07	2026-03	Laptop	South	3	55000	4500
9	ORD010	2026-03-15	2026-03	Keyboard	West	20	1500	1000
10	ORD011	2026-03-21	2026-03	Headphones	North	8	2500	400
11	ORD012	2026-04-04	2026-04	Laptop	East	2	55000	2500
12	ORD013	2026-04-11	2026-04	Smartphone	West	10	22000	4000
13	ORD014	2026-04-19	2026-04	Keyboard	South	15	1500	750
14	ORD015	2026-04-25	2026-04	Headphones	East	12	2500	600

2. **Column names:**

[2]:
```
df.columns
```

[2]:
```
Index(['order_id', 'order_date', 'month', 'products', 'region', 'units',
       'unitprice', 'discount'],
      dtype='str')
```

3. **Count products sold by order frequency:**

```
[3]: df["products"].value_counts()
```

```
[3]: products
     Laptop        4
     Smartphone    4
     Headphones    4
     Keyboard      3
     Name: count, dtype: int64
```

4. **Calculate revenue by region**

```
[4]: df["revenue"] = df["units"] * df["unitprice"] - df["discount"]

     df.groupby("region")["revenue"].sum()
```

```
[4]: region
     East     291600
     North    145500
     South    463250
     West     335750
     Name: revenue, dtype: int64
```

5. **Region summary – mean, order count and total revenue**

```
[5]: region_summary = df.groupby("region").agg(
         average_revenue=("revenue", "mean"),
         order_count=("order_id", "count"),
         total_revenue=("revenue", "sum")
     )

     region_summary.sort_values(
         by="total_revenue", ascending=False
     )
```

[5]:

region	average_revenue	order_count	total_revenue
South	115812.5	4	463250
West	83937.5	4	335750
East	72900.0	4	291600
North	48500.0	3	145500

6. **Calculate units sold by product**

```
[6]: df.groupby("products")["units"].sum().sort_values(ascending=False)
```

```
[6]: products
     Keyboard      47
     Headphones    45
     Smartphone    29
     Laptop         8
     Name: units, dtype: int64
```

7. **Calculate correlation table**

```
[7]: numeric_columns = df.select_dtypes(include="number").columns

     c_table = df[numeric_columns].corr()

     display(c_table)
```

	units	unitprice	discount	revenue
units	1.000000	-0.857995	-0.423685	-0.464018
unitprice	-0.857995	1.000000	0.606274	0.543580
discount	-0.423685	0.606274	1.000000	0.925685
revenue	-0.464018	0.543580	0.925685	1.000000

8. **Calculate units and revenue correlation**

```
[8]: df["units"].corr(df["revenue"])
```

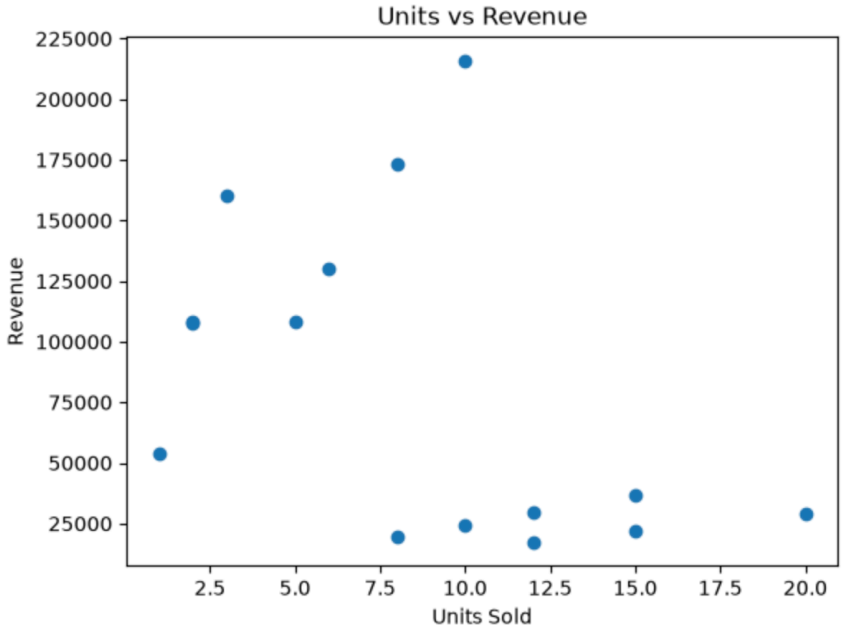
```
[8]: np.float64(-0.464018148503504)
```

9. Scatter plot – units vs revenue

[9]:
```
plt.scatter(df["units"], df["revenue"])

plt.xlabel("Units Sold")
plt.ylabel("Revenue")
plt.title("Units vs Revenue")

plt.show()
```



10. Calculate monthly revenue using groupby

[10]:
```
monthly_revenue = df.groupby("month")["revenue"].sum().sort_index()

display(monthly_revenue)
```

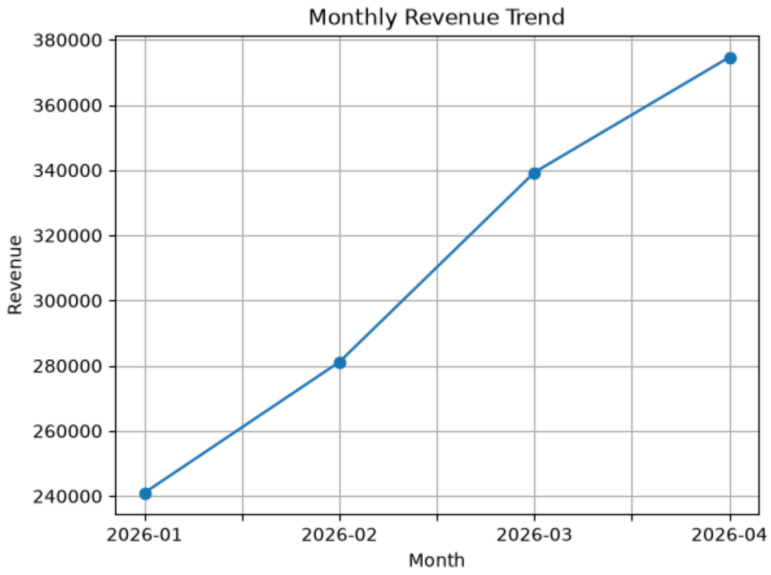
```
month
2026-01    241000
2026-02    281150
2026-03    339300
2026-04    374650
Name: revenue, dtype: int64
```

11. Line plot – month vs revenue

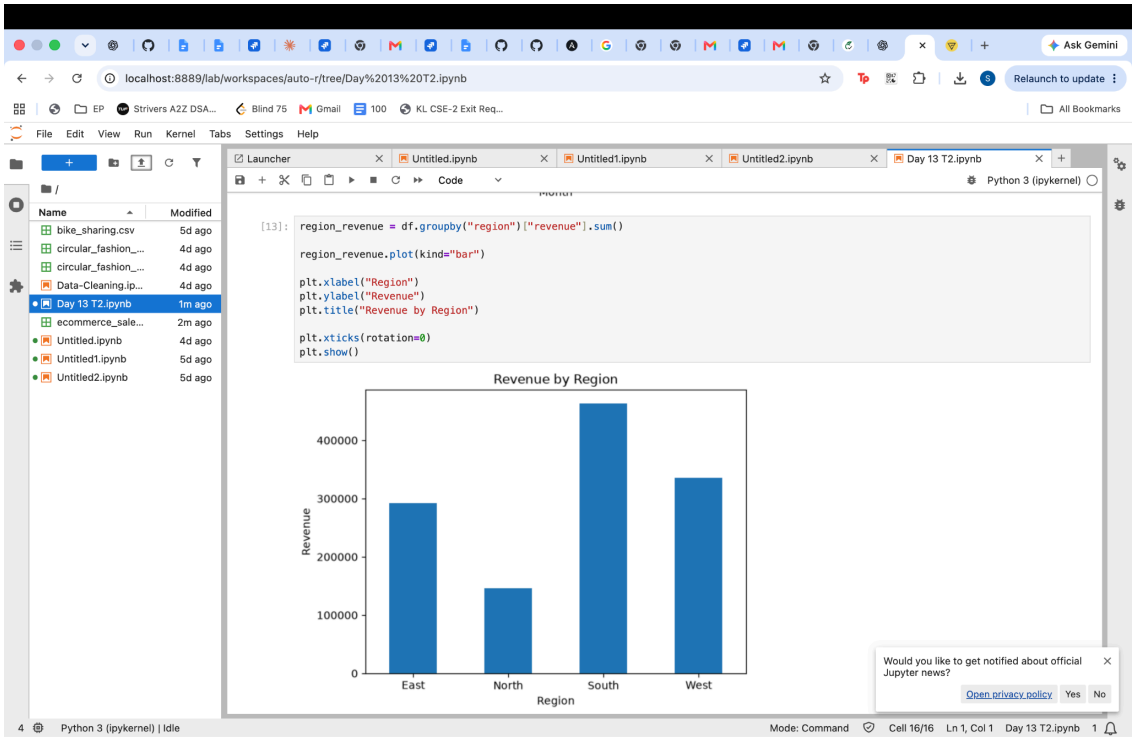
[12]:
```
monthly_revenue.plot(kind="line", marker="o")

plt.xlabel("Month")
plt.ylabel("Revenue")
plt.title("Monthly Revenue Trend")

plt.grid(True)
plt.show()
```



12. Bar chart – revenue by region



13. Find highest revenue orders using sorting

[14]:
```
highest_revenue = df.sort_values(
    by="revenue", ascending=False
)

display(highest_revenue)
```

	order_id	order_date	month	products	region	units	unitprice	discount	revenue
12	ORD013	2026-04-11	2026-04	Smartphone	West	10	22000	4000	216000
4	ORD005	2026-02-09	2026-02	Smartphone	South	8	22000	3000	173000
8	ORD009	2026-03-07	2026-03	Laptop	South	3	55000	4500	160500
7	ORD008	2026-03-02	2026-03	Smartphone	East	6	22000	1800	130200
1	ORD002	2026-01-08	2026-01	Smartphone	North	5	22000	1500	108500
0	ORD001	2026-01-05	2026-01	Laptop	South	2	55000	2000	108000
11	ORD012	2026-04-04	2026-04	Laptop	East	2	55000	2500	107500
3	ORD004	2026-02-03	2026-02	Laptop	West	1	55000	1000	54000
6	ORD007	2026-02-20	2026-02	Headphones	West	15	2500	750	36750
14	ORD015	2026-04-25	2026-04	Headphones	East	12	2500	600	29400
9	ORD010	2026-03-15	2026-03	Keyboard	West	20	1500	1000	29000
2	ORD003	2026-01-12	2026-01	Headphones	East	10	2500	500	24500
13	ORD014	2026-04-19	2026-04	Keyboard	South	15	1500	750	21750
10	ORD011	2026-03-21	2026-03	Headphones	North	8	2500	400	19600
5	ORD006	2026-02-14	2026-02	Keyboard	North	12	1500	600	17400

14. Display top 5 highest revenue orders

[15]:
```
df.nlargest(5, "revenue")
```

[15]:

	order_id	order_date	month	products	region	units	unitprice	discount	revenue
12	ORD013	2026-04-11	2026-04	Smartphone	West	10	22000	4000	216000
4	ORD005	2026-02-09	2026-02	Smartphone	South	8	22000	3000	173000
8	ORD009	2026-03-07	2026-03	Laptop	South	3	55000	4500	160500
7	ORD008	2026-03-02	2026-03	Smartphone	East	6	22000	1800	130200
1	ORD002	2026-01-08	2026-01	Smartphone	North	5	22000	1500	108500